

Lecture 17.2: Chomsky Normal Form (Examples)

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Examples on converting Context Free Grammar to Chomsky Normal Form:

1. Example 1: $\Sigma = \{a, b\}$

$S \rightarrow ASB \mid a$

$A \rightarrow aAS \mid a \mid \epsilon$

$B \rightarrow SbS \mid A \mid bb$

Step 1: Add a new start state.

$S' \rightarrow S$

$S \rightarrow ASB \mid a$

$A \rightarrow aAS \mid a \mid \epsilon$

$B \rightarrow SbS \mid A \mid bb$

Step 2: Remove ϵ -productions

$S' \rightarrow S$

$S \rightarrow ASB \mid a \mid AS \mid SB$

$A \rightarrow aAS \mid a \mid aS$

$B \rightarrow SbS \mid A \mid bb$

Step 3: Remove Unit Productions

$S' \rightarrow ASB \mid a \mid AS \mid SB$

$S \rightarrow ASB \mid a \mid AS \mid SB$

$A \rightarrow aAS \mid a \mid aS$

$B \rightarrow SbS \mid aAS \mid a \mid aS \mid bb$

Step 4: Make CNF form

$S' \rightarrow XB \mid a \mid AS \mid SB$

$S \rightarrow XB \mid a \mid AS \mid SB$

$A \rightarrow YX \mid a \mid YS$

$B \rightarrow SZS \mid YX \mid a \mid YS \mid ZZ$

$X \rightarrow AS$

$Y \rightarrow a$

$Z \rightarrow b$

Step 5: Complete

$S' \rightarrow XB \mid a \mid AS \mid SB$

$S \rightarrow XB \mid a \mid AS \mid SB$

$A \rightarrow YX \mid a \mid YS$

$B \rightarrow WS \mid YX \mid a \mid YS \mid ZZ$

$X \rightarrow AS$
 $Y \rightarrow a$
 $Z \rightarrow b$
 $W \rightarrow SZ$

2. Example 2: $\Sigma = \{a, b\}$

$S \rightarrow ASA \mid aB$
 $A \rightarrow B \mid S$
 $B \rightarrow b \mid \epsilon$

Step 1:

$S' \rightarrow S$
 $S \rightarrow ASA \mid aB$
 $A \rightarrow B \mid S$
 $B \rightarrow b \mid \epsilon$

Step 2:

$S' \rightarrow S$
 $S \rightarrow ASA \mid aB \mid AS \mid SA \mid a$
 $A \rightarrow B \mid S$
 $B \rightarrow b$

Step 3:

$S' \rightarrow ASA \mid aB \mid AS \mid SA \mid a$
 $S \rightarrow ASA \mid aB \mid AS \mid SA \mid a$
 $A \rightarrow b \mid ASA \mid aB \mid AS \mid SA \mid a$
 $B \rightarrow b$

Step 4:

$S' \rightarrow YA \mid XB \mid AS \mid SA \mid a$
 $S \rightarrow YA \mid XB \mid AS \mid SA \mid a$
 $A \rightarrow b \mid YA \mid XB \mid AS \mid SA \mid a$
 $B \rightarrow b$
 $X \rightarrow a$
 $Y \rightarrow AS$

3. Example 3: $\Sigma = \{a, b, c\}$

$S \rightarrow aXbX$

$X \rightarrow aY \mid bY \mid \epsilon$

$Y \rightarrow X \mid c$

Do it yourself.

Normal form of a context free grammar makes the algorithm for membership problem (whether a string belongs to the Language of the grammar or not) to be run in Polynomial time.